

Machine Learning Technology Readiness Level (MLTRL) Full Assessment Report & Cards as of July 22, 2026

1. System-Level MLTRL Card

TECHNOLOGY NAME	ACCESS-DP Mobile Health Caregiver AI Support Application	CURRENT MLTRL = TRL 4	
TL;DR SUMMARY	An offline-first mobile clinical decision support suite for informal and formal caregivers. Combines three interconnected decision engines: (1) UC2 real-time telemetry anomaly scoring via an on-device 12D TFLite Dense Autoencoder, (2) UC3 longitudinal rehab trajectory failure tracking with sigmoid priority scoring, and (3) UC4 personalized caregiver micro-priority checklist scoring. Executed locally within a React Native client using encrypted SQLite persistence (SQLCipher) and hard-coded emergency safety overrides.		
MODEL & ALG DETAILS	• UC2 Anomaly Engine: 12-Dimensional Dense Autoencoder (tiny_uc2_autoencoder12.tflite) with reconstruction MSE threshold $\tau = 1.1447161$ (97.5th percentile). • UC3 Rehab Engine: Rule-guided hybrid decision model with deterministic thresholds and calibrated logistic sigmoid priority scoring: $z = -2.2 + 7.5*(romGap) + 0.32*(plateauDays) + 2.0*\max(0, adherence - 0.80) + 0.10*\max(0, pain - 3.0) + 0.10*\max(0, fatigue - 4.0) + 1.2*(walkingGap) + 0.12*(complexityScore)$; $PriorityScore = \text{Sigmoid}(z)$. • UC4 Care Focus Engine: 8-stage safety-bounded rule engine with clamped scoring formula: $Final\ Score = Clamp_0_1(RuleScore + BlindSpotBonus + UsefulnessBonus - RepeatPenalty - DismissPenalty)$.		
METRICS & RESULTS	• UC2 Validation: ROC-AUC = 0.9784, Event Detection Recall = 0.8678, F1-Score = 0.9057, Normal FPR = 0.029 (N=632 normal / N=764 anomaly). 100% logic parity across fixture datasets. • UC3 Validation: Trajectory Failure Detection Recall = 0.9473, Precision = 0.9236, Specificity = 0.9730. Safety Gate Override Primacy = 100% • UC4 Validation: 100% test pass rate on validateUC4Registries) and frozen JSON validation reports. Blind-spot card surface rate = 0.8420.		
DATA CONSIDERATIONS	Local data persistence managed via encrypted SQLite database (SQLCipher) on React Native mobile client. Processes watch telemetry streams, FHIR R4 resource representations (Patient, Observation, CarePlan, CommunicationRequest, DocumentReference), historical care logs, and baseline EHR complexity profiles. Datasets used for Model training and validation: • MIMIC-IV (Johnson et al.) & MMASH (Rosse et al.)		
ETHICS & SAFETY	Hard emergency rule overrides (e.g., $SpO_2 \leq 88\%$, $HR \geq 140$ bpm, acute Red Flag symptoms, UC1/UC2 Severity 3 active emergency) operate on an insuppressible fast-path that bypasses ML inference completely. Prohibited from making diagnostic claims or drug causality claims. Operates strictly under human caregiver oversight.		
CAVEATS & EDGE CASES	Mobile application wrappers currently have static EHR fallback defaults. Rule thresholds rely on clinician configuration overrides for patient-specific tuning. Live background ingestion latency management is in progress on physical hardware.		
KEY ASSUMPTIONS	1. Telemetry input vectors undergo pre-validation for non-physiologic rate-of-change jumps. 2. Missing vital features and clinical parameters are imputed using EHR profile baselines or population medians. 3. Medically excused exercise skips preserve full adherence credit to prevent unfair non-compliance labeling. 4. Caregiver HITL feedback is submitted via structured observation matrices and response schemas.		
INTENDED USE	Clinical decision support and real-time monitoring tool for informal and formal caregivers managing pediatric and adult patients in home health environments. Operates as a decision-support checklist and trajectory monitor requiring human caregiver oversight.		

2. (For Appendix): Detailed Subsystem MLTRL Assessment Cards

A.1 UC2 Health Anomaly & Recurrence Decision Engine (TRL 5)

TECHNOLOGY NAME	UC2 AI Health Anomaly Detection Engine	CURRENT MLTRL = TRL 5	
TL;DR SUMMARY	A 10-step offline-first decision pipeline combining rate-of-change artifact pre-validation, deterministic emergency		

	rules, on-device 12D TFLite autoencoder anomaly scoring ($\tau = 1.1447161$), a 13x7 caregiver Human-In-The-Loop (HITL) matrix, and SQLite historical look-back recurrence scoring. Integrated software pipeline validated against offline dataset fixtures, with live HealthKit streaming and HIL synthetic anomaly paths verified on physical hardware.
MODEL & ALG DETAILS	• Model Architecture: 12-Dimensional Dense Autoencoder • Input Vector (12D): Watch-native features (Heart Rate, SpO2, Respiratory Rate, HRV SDNN, Body Temp, Activity, Steps, Calories, Sleep Quality, hour_sin, hour_cos, is_sleep_window) • Anomaly Scoring: Reconstruction MSE evaluated against $\tau = 1.1447161$ (97.5th percentile validation MSE). Top feature contributors dynamically attributed upon anomaly detection.
METRICS & RESULTS	• Offline Model Validation: ROC-AUC = 0.9784, Event Detection Recall = 86.78%, F1-Score = 0.9057, Normal False Positive Rate = 2.85% (N=632 normal / N=764 anomaly samples). • Pipeline Logic Parity: 100% test pass rate across fixture datasets, confirming deterministic parity across rule engines, scaler transformations, and decision routing.
DATA CONSIDERATIONS	Utilizes curated watch telemetry datasets, synthetic edge-case injections, and FHIR/EHR profile baselines. Local data persistence is managed via an encrypted SQLite store for historical 48h/72h recurrence look-back windows. Datasets used for Model training and validation: • MIMIC-IV (Johnson et al.) • MMASH (Rosse et al.)
ETHICS & SAFETY	Hard emergency overrides (e.g. $\leq 88\%$ SpO2, ≥ 140 bpm HR) operate on a fast-path that bypasses ML inference and strictly cannot be overridden or suppressed by patient baselines.
CAVEATS & EDGE CASES	On-device physical hardware validation uses native TFLite bindings. Recurrence risk scoring is capped at Severity Floor 2 for non-emergencies. On-device streaming latency and processing overhead are currently being optimized.
KEY ASSUMPTIONS	1. Telemetry input vectors undergo pre-validation for non-physiologic rate-of-change jumps. 2. Missing vital features are imputed using EHR profile baselines. 3. Caregiver HITL feedback is submitted via structured 13x7 observation matrices.
INTENDED USE	Clinical decision support and real-time monitoring tool for caregivers and healthcare providers monitoring pediatric and adult patients in home health environments. Requires human-in-the-loop oversight.

A.2 UC3 Rehabilitation Trajectory & Failure Decision Engine (TRL 4)

TECHNOLOGY NAME	UC3 Rehabilitation Trajectory & Failure Decision Engine	CURRENT MLTRL = TRL 4	
TL;DR SUMMARY	Executes multi-metric longitudinal home rehabilitation tracking through a six-stage deterministic and rule-guided pipeline: (1) Adherence Derivation, (2) Data Quality Assessor, (3) Hard Safety Gate, (4) Longitudinal Trajectory Analysis, (5) Baseline EHR Complexity Scoring, and (6) Review Priority Logistic Scoring. Operates completely offline-first within React Native using local SQLite database storage.		
MODEL & ALG DETAILS	• Architecture: Rule-guided hybrid decision model with deterministic thresholds and calibrated logistic sigmoid scoring. • Tracked Metrics (6D): Range of Motion (romDegrees), Active Repetitions (exerciseReps), Compliance Ratio (adherence), Pain Level (painScore), Fatigue Level (fatigueScore), and Ambulation (walkingMinutes). • Priority Scoring Formula: $z = -2.2 + 7.5*(romGap) + 0.32*(plateauDays) + 2.0*\max(0, adherence - 0.80) + 0.10*\max(0, pain - 3.0) + 0.10*\max(0, fatigue - 4.0) + 1.2*(walkingGap) + 0.12*(complexityScore)$$PriorityScore = \text{Sigmoid}(z) = 1 / (1 + e^{(-z)})$		
METRICS & RESULTS	• Deterministic Verification: Unit test suites verify exact trajectory, plateau, and priority score calculations across 100% of scenario test cases. Runtime and storage integrity verified on React Native SQLite client . Trajectory Failure Detection Recall = 0.9473, Precision = 0.9236, Specificity = 0.9730. Safety Gate Override Primacy = 100%		
DATA CONSIDERATIONS	Exports compliant JSON-RPC 2.0 payloads with fhirR4Hint specifying resource types (Patient, Observation, CarePlan, CommunicationRequest, DocumentReference). Local data persistence uses encrypted SQLite storage (SQLCipher) and native keystore integration.		
ETHICS & SAFETY	Acute red-flag safety rules execute prior to trajectory math, overriding non-emergency logic. Medically excused skipped days (documented illness or clinician stop orders) preserve full adherence credit to protect patients from unfair non-compliance labeling. Produces distinct caregiver-facing messages and clinician-facing summaries.		

CAVEATS & EDGE CASES	Current implementation uses fallback static EHR defaults when dynamic EHR context is omitted. Rule thresholds rely on clinician configuration overrides for patient-specific tuning.
KEY ASSUMPTIONS	1. Logs with medical skip reasons (fever, fall, acute pain, doctor directive) automatically assign 100% adherence credit. 2. EHR complexity scores are calculated transparently across clinical history, mobility limitations, and safety considerations. 3. Baseline rehabilitation milestone curves are accurately defined by physical therapy care plans.
INTENDED USE	On-device real-time rehab tracking and clinical decision support tool for caregivers and physical therapists to identify post-acute rehabilitation trajectory failure and escalation risks. Must operate with human clinical oversight.

A.3 UC4 Adaptive Care Focus Checklist & Personalized Priorities Engine (TRL 4)

TECHNOLOGY NAME	UC4 Adaptive Care Focus Checklist Engine	CURRENT ML-TRL = TRL 4	
TL;DR SUMMARY	Executes deterministic, patient-specific caregiver micro-priority selection through an 8-stage safety-bounded pipeline: (1) active emergency pause interception, (2) context aggregation, (3) declarative rule evaluation, (4) normalized candidate scoring, (5) domain-based diversity selection (max 3 cards), (6) template hydration, (7) audit logging, and (8) offline local SQLite persistence.		
MODEL & ALG DETAILS	• Rule Engine: ACCESS-DP Micro-Priority Rule Registry • Clamped Mathematical Formula: Final Score = Clamp_0_1(RuleScore + BlindSpotBonus + UsefulnessBonus - RepeatPenalty - DismissPenalty) Where: RuleScore = Clamp_0_1(Sum of fired rule weights); RepeatPenalty = Min(0.18, RecentSameTemplateCount * 0.06); DismissPenalty = Min(0.24, DismissedOrNotRelevantCount * 0.12); UsefulnessBonus = Min(0.12, HelpfulOrLoggedCount * 0.06); BlindSpotBonus = 0.08 (if template belongs to unshown blind-spot domain).		
METRICS & RESULTS	• Deterministic Algorithmic Validation: Proof of concept validated via generating frozen JSON validation reports with 100% test pass rate. Emergency Intercept Gate = 100% pass rate. Blind-spot Domain Card Surface Rate = 84.2%		
DATA CONSIDERATIONS	Converts raw event history into normalized context features using controlled observation and context codes. Stores execution state locally in React Native SQLite DB. Emits typed UC4AuditRecord events for full line-item auditability.		
ETHICS & SAFETY	Hardcoded template safety tags (no_diagnosis) restrict output to observation prompts. Medication rules map to watch areas for timing context tracking only; prohibited from claiming drug causality. Immediate execution pause when UC1/UC2 active emergency is detected . Free-text fields isolated from scoring (usedForScoring: false). SLM/LLMs strictly prohibited from scoring/selection.		
KEY ASSUMPTIONS	1. Pure logical rule functions return structured evidence references and base weights. 2. Candidate card rendering is enforced with domain diversity selection capped at max 3 cards. 3. LLMs are restricted post-hoc to text polishing after cards are locked.		
INTENDED USE	On-device, real-time decision support tool for informal and formal caregivers to highlight patient-specific care focus areas and capture structured follow-up observations. Operates strictly under human caregiver oversight as a decision-support checklist.		

3. (For Appendix): Cross-Engine Verification & Validation (V&V) Matrix

Engine / Area	Target Requirement & Criterion	Verification Method & Evidence	Status & TRL
UC2 Determinism	100% pipeline pass rate on fixture dataset test suites (parity.ts) across all 10 modules.	Automated fixture parity test execution.	VERIFIED
UC2 Emergency Override	Hard emergency rules bypass ML inference with 0 false negatives.	Boundary testing and HIL synthetic anomaly injection on physical devices.	VERIFIED
UC2 Edge Execution	Execute full decision engine on physical devices using native TFLite runtime bindings.	On-device physical hardware validation using TFLite C++ interpreter bindings.	VERIFIED
UC3 Safety Primacy	Acute Red Flag symptoms trigger immediate escalation with Priority 1.0.	Red Flag symptom injection test suites overriding trajectory math.	VERIFIED

Engine / Area	Target Requirement & Criterion	Verification Method & Evidence	Status & TRL
UC3 Medical Exception	Preserve adherence credit during documented medical illness pauses.	Log evaluation test cases confirming 100% compliance credit during illness.	VERIFIED
UC4 Registry Integrity	All rule weights in [0,1], valid schema mappings, no free-text scoring.	Execution of validateUC4Registries() and frozen JSON validation reports.	VERIFIED
UC4 Emergency Pause	Zero routine cards rendered when emergency context triggers.	Elena emergency pause payload test cases confirming immediate execution halt.	VERIFIED